

Individual Work Breakdown Report

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Project Context

This document presents my individual contribution to the team project “Dublin Housing Affordability Analysis (2015–2025).” The project required integration of multi-source datasets, end-to-end ETL implementation, analysis, and communication through static and interactive visual outputs.

Primary Ownership Areas

- Exploratory analysis and visual design.
- Streamlit dashboard implementation and refinement.
- Results interpretation and insight articulation.
- Presentation narrative coordination.

Detailed Contributions Across Project Stages

1. Analytical Framing and Visual Planning

I helped define the visual questions that the project needed to answer:

- How have price and income moved over time?
- How does affordability evolve where income overlap exists?
- Which areas show consistently high prices?
- Are area-level patterns persistent or short-term?

This framing guided chart selection and dashboard sequence.

2. Static Analysis Outputs

I contributed to chart generation logic for trend and area analyses, ensuring consistency between transformed metrics and displayed visual encodings. I also reviewed axis labels, titles, and readability to maintain clarity for non-technical audiences.

3. Dashboard Development

I implemented and refined the Streamlit dashboard structure:

1. KPI cards for headline values,
2. city-level trend chart,
3. affordability trend chart,
4. year-driven area comparison,
5. area-year heatmap,
6. key insight text summary.

I improved usability by maintaining concise interaction controls and balancing detail with readability.

4. Visualization Quality Improvements

I participated in resolving several quality issues:

- corrected area-label issues caused by upstream locality extraction limits,
- adjusted affordability plotting to avoid misleading continuation beyond valid overlap years,
- improved chart semantics by removing confusing color mappings and moving volume context to hover data.

These updates significantly improved interpretability and credibility of findings.

5. Results Interpretation and Insight Development

I led drafting of key findings in the report and dashboard:

- persistent divergence between price and income growth,
- affordability deterioration over overlap years,
- structural area-level inequality across time.

I ensured insights were supported by visual evidence and aligned with computed metrics.

6. Communication and Presentation

I contributed to presentation structure and speaking-flow design, especially for sections explaining how visual outputs connect to the research question and policy interpretation. I also supported preparation of concise, audience-appropriate explanations for viva discussion.

Cross-Stage Evidence of Participation

- **Selection:** participated in source relevance and variable mapping discussion.
- **Collection/Storage:** validated that curated outputs were dashboard-ready.
- **Processing:** reviewed transformed features for analytical use.
- **Analysis:** primary owner of interpretation and result narrative.
- **Visualization:** primary owner of dashboard composition and clarity.
- **Communication:** co-led final storytelling for report and presentation.

Challenges and Responses

Challenge 1: Translating complex outputs into simple visuals without losing analytical meaning. **Response:** Used progressive narrative layout and limited interactions to essential controls.

Challenge 2: Mismatch between data coverage and visual continuity. **Response:** Restricted affordability chart to valid overlap years and clarified metric context.

Challenge 3: Spatial labels initially lacking interpretability. **Response:** Coordinated with transformation updates and verified improved area representation in charts.

Reflection

My contribution focused on making analysis understandable and decision-useful. Effective analytics communication depends on both numerical correctness and visual trustworthiness. By refining chart design and dashboard flow, I helped ensure the project's findings could be interpreted accurately by technical and non-technical stakeholders.

Declaration

This report is an original account of my own contribution to the team project.